

Tremor Stabilization for Sculpting Assistance in Virtual Reality

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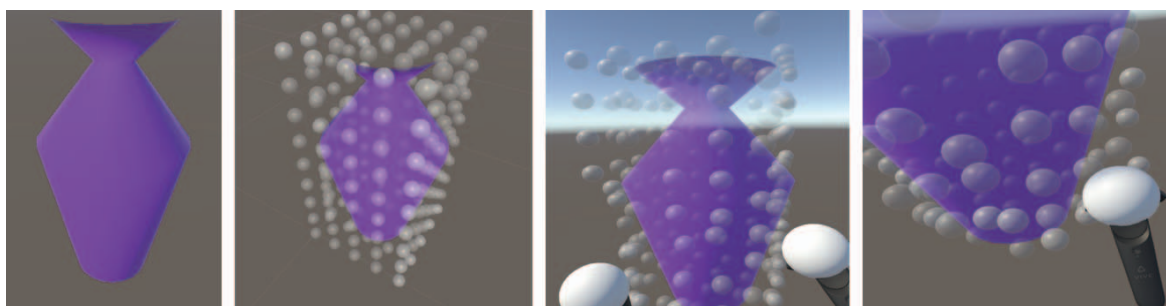


Figure 1: Images of our tremor stabilization interface for a sculpting application, in which a user pushes the gray spheres to match the shape of the vase, allowing us to measure the distance between the target object and the user's creation. These show a target vase (left), a rectangular solid shape consisting of manipulable spherical building blocks (middle left), the user pushing these spheres into place (middle right), and the process nearing completion (right).

ABSTRACT

This paper presents an exploration of assistive technology for virtual reality (VR) art, such as sculpting and ceramics. For many artists, tremors from Parkinsonian diseases can interfere with molding, carving, cutting, and modeling of different mediums for creating new sculptures. To help address this, we have developed a system that algorithmically stabilizes tremors to enhance the artistic experience for creators with physical impairments or movement disorder. In addition, we present a real-time sculpting application that allows us to measure differences in sculpting actions and a target object or shape.

Keywords: assistive interface, sculpting, virtual reality, tremors, art

1 INTRODUCTION AND RELATED WORK

Virtual Reality (VR) offers a unique opportunity to develop methods that facilitate the exploration of environments and experiences that would typically be less accessible or more challenging to engage with. One such underexplored application of VR is ceramics, particularly focusing on individuals with physical impairments that would prevent them from partaking in conventional, non-VR ceramics settings. In this paper, we have developed a method to stabilize the oscillatory movements characteristic of the hand tremors resulting from Parkinson's Disease.

The idea of stabilizing tremors for human computer interfaces dates back several decades, including algorithms to reduce tremor for microsurgery [5]. Stabilization has also been used in prior research for techniques like the virtual hand [6] and for general interaction [1], but it has not yet been applied to fields like art, sculpting, or writing in VR. Other research has targeted movement coordination disorders such as dyspraxia, though this is largely limited to developmental training and does not scale well to systemic diseases such as Parkinson's [3]. More recent work has even sought to reduce tremor in teleoperation or telepresence applications by using convolutional

neural networks [4]. Moreover, there is a significant lack of other assistive technologies that help improve accessibility to immersive technologies [2]. Our objective is to help close this gap.

More specifically, the goal of this project is to stabilize the motion of VR users with Parkinson's Disease or related movement disorders and improve the fine-grained accuracy of manipulation of objects in the scene. Our method allows for stabilized sculpting of the contours of virtual objects, one of which is shown in Figure 1. Our objective is to allow a participant with Parkinson's Disease to create a target form in the scene more quickly than they would without stabilization. The interface also provides a testing arena where a user can manipulate spheres into place to match a target sculpture.

2 ENVIRONMENT SETUP AND SCULPTURE MANIPULATION

The first part of our system is an interactive environment that lets users create sculptures by manipulating basic building blocks (in our case, spheres) in the scene. To accomplish this, we first programmatically generate a target object mesh for the user to practice with, such as a vase, as shown in the leftmost image of Figure 1. Rather than creating a form by starting with an existing mesh, the user can displace individual blocks that are formed into a rectangular solid that encapsulates the practice mesh. This takes the form of a three-dimensional grid of cube game objects of consistent size and spacing, as shown in the second image in Figure 1.

Within the bounds of this grid is the vase-shaped semi-transparent target mesh. Using a controller, the user is able to manipulate the position of the spheres by pressing against them in the desired direction of manipulation. The primary objective of this interface is to have users align the spheres with the boundary of the target mesh and measure the differences between the interface with and without stabilization. Displacement of the individual spheres is achieved by repeatedly adding a fixed increment to their position as long as the bounding boxes of the spheres and controller pointers are in a collision state. This method ensures a controlled and uniform adjustment of the spheres' positions during the interaction phase.

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3 STABILIZATION ALGORITHM

Next, we needed to design a method that would reduce perceived tremors in the virtual environment. To do so, we wrote an algorithm that would smooth the motion of the controller such that tremors were reduced, but also so that the sculpting pointer was still responsive enough to make manipulations. Since the frequency of Parkinson's tremors generally falls between 4Hz and 6Hz, we designed a filter that averaged the position of frames over a large enough number of frames to ensure multiple tremor cycles were captured in the filter. Though mathematically, we needed at least 10 frames for this to be the case, we ended up using 20 frames through logically testing the interface while shaking our hands at roughly 6 cycles per second.

This filtered position is then applied to the controller pointer, and the controller renderer is disabled (but is shown for reference in Figure 1), which prevents the user from seeing both the oscillating controller and stabilized pointer at the same time. Moreover, the pointer position is translated up and away from the center of the hand, as if it were in front of the user's extended index finger. This was done to mimic the position of a sculpting or carving tool, which generally extends past the hand. In addition to smoothing the position of the tool for manipulation, the participant also has a more stable visual guide with which to manipulate a target model. A sample output of this stabilization is shown in Figure 2.

4 INITIAL TESTING AND REFINEMENT

To test our stabilization tool, we sculpted a test model to see how well we could manipulate spheres into the target positions specified by the vase. The resulting process and model is shown in Figure 1. Initially, the objects rendered to sculpt the vase were cubes, but one modification we had to make was to change these to spheres. Perceptually, we expected that any perturbation of the cube would happen in-line with the axis corresponding to the face of the cube that is pressed. For example, if we pressed on the right x face, we expected that it would be translated to the left, regardless of where on the face we pressed. This turned out to be very inefficient for making minute movements in a direction relative to the cube on multiple axes, i.e., those that might have combined x, y, and z components.

To solve this problem, we switched the cubes to spheres, which provided a more intuitive guide to making multiple-axis movements. Because any intersection with a sphere would cause movement orthogonal to the point of intersection in the direction of the sphere center, it was now possible to move the sphere in any direction without having to conduct multiple, single-axis movements.

Finally, we also required the trigger to be pressed in order to activate manipulations. Though leaving the tool on constantly might be useful for large-area surface culling, it prevented the user from putting the tool through to the middle of the object to push elements outwards, i.e., inside-out extrusion. In future evaluations, final positions of the set of cubes could be compared to the boundary of the target mesh to evaluate its accuracy. We have this planned as future work.

5 DISCUSSION AND CONCLUSION

The initial results from testing the tremor stabilization algorithm in VR is encouraging, particularly in demonstrating the feasibility of using this technique to assist artists with Parkinson's Disease or other similar movement disorders.

The implementation of the tremor stabilization algorithm within our VR system showed promising results in initial testing. Specifically, the smoothing effect of the algorithm led to a significant reduction in erratic movements caused by perturbations, allowing for more accurate movement of the cubes and alignment with the mesh. While our stabilization method works, the system as a whole needs extensive testing with a wider range of participants of differing impairment levels to identify weaknesses and make improvements

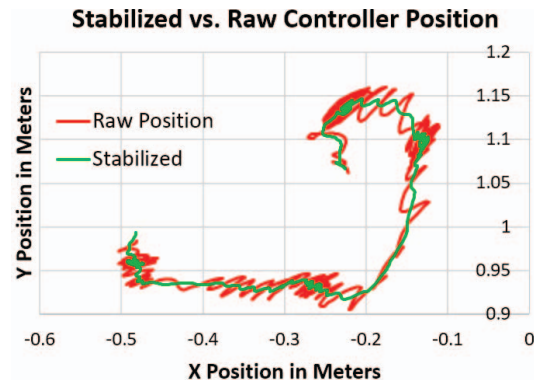


Figure 2: Scatter plot of raw vs. stabilized controller positions. The red line shows how tremors might affect a sculptor's hand positions, and the green line demonstrates what the stabilized trajectory would look like.

to the system. Areas such as user interface design, responsiveness of the environment to subtle hand movements, and long-term user comfort should be a priority moving forward.

We are currently working to incorporate a tremor rate calibration function with the goal of personalizing the experience based on the tremor characteristics of the artist, thereby enhancing the system's adaptability and effectiveness. This would also allow users with other tremor disorders to use the system by adjusting the responsiveness of the controller in accordance with the frequency of the most common hand tremors. Lastly, we would like to incorporate the ability to 3D print the sculptures created within this VR system. This will bridge the gap between virtual creations and physical art, allowing artists to bring their digital creations into the tangible world. These pieces can then be glazed and kiln-fired in the traditional way.

Through these advancements, we hope to not only improve the VR artistic experience for individuals with tremor disorders but also to inspire further research and development in adaptive technologies across various fields. We hope this will be a step towards more inclusive and accessible VR-based environments, catering to individuals with physical impairments.

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